8 **C** $P_D = F_D \times v$

$$\Rightarrow F_D = \frac{30\times10^3}{25} = 1200 \text{ N}$$

Since it is moving at constant speed, $F_R = F_D \Rightarrow F_R = 1200$ N

On the slope, $F_D = mg\sin\theta + F_R$ since moving at constant speed

$$\Rightarrow F_D = 1400\times9.81\times\sin 2^o + 1200 = 1679 \text{ N}$$

$$P_D = F_D \times v = 1679\times25 = 42 \text{ kW}$$